

AI-GUIDED COMBINATION THERAPY DISCOVERY FOR TRIPLE-NEGATIVE BREAST CANCER

A Multi-Agent Computational Pipeline with Wet-Lab Validation

Research Proposal

Computational Oncology • Precision Medicine • Multi-Agent AI

Version 1.1 • July 2026

Specific Aims

Triple-negative breast cancer (TNBC) accounts for 15–20% of breast cancers and lacks the targeted single-agent options available for hormone-receptor-positive or HER2-amplified disease, making rationally-designed combination therapy a central strategy rather than an afterthought. Selecting which combination fits a given patient's genomic profile, however, remains largely a matter of expert judgment applied to an intractably large combinatorial space.

This proposal describes a working, tested computational pipeline (“MC-ORE / Hybrid-CORE”) that generates patient-specific combination hypotheses directly from sequencing data, and a wet-lab validation plan to test its output against real biology. The pipeline has already been implemented, run on real TCGA-BRCA data, and used to correct its own methodological errors — this is a proposal to validate and extend a working prototype, not a purely conceptual system.

- **Aim 1:** Develop and validate a multi-agent computational pipeline that identifies patient-specific multi-drug combinations from real genomic data, integrating knowledge-graph reasoning, a synergy-scoring model, and a combinatorial regimen search (MDCOE) capable of proposing 2–5-drug regimens rather than pairs alone.
- **Aim 2:** Experimentally validate model-generated hypotheses in a confirmed-TNBC patient sample via a two-arm wet-lab pilot testing two mechanistically distinct combinations derived from the same tumour's genomic profile.
- **Aim 3:** Close the loop between wet-lab and model by feeding synergy, toxicity, and clinician-acceptance data back into the pipeline's knowledge graph and scoring functions, using early results to correct at least one identified scoring bias before scaling to additional patients.

Significance and Background

TNBC is defined by the *absence* of the three most common breast-cancer drug targets (ER, PR, HER2 amplification), which is precisely why it disproportionately relies on combination chemotherapy and, increasingly, genomically-guided targeted combinations. TP53 is altered in over 80% of TNBC, making it the single most consequential gene in this disease — but as this proposal's preliminary data show, “TP53 altered” is not one biological state: a missense mutant, a nonsense-mediated-decay-degraded truncating mutant, and a biallelic deletion each call for a different drug strategy, a distinction most computational pipelines (including an earlier version of this one) do not make.

Beyond TP53, TNBC tumours frequently carry alterations across the receptor tyrosine kinase (RTK) and non-receptor tyrosine kinase (NRTK) families — 58 and 32 genes respectively, per the standard Robinson/Wu/Lin kinome classification — many with real approved or investigational inhibitors, but scattered across a search space too large for manual triage.

Innovation

- **Search beyond pairs.** MDCOE, the regimen-search layer developed for this proposal, uses beam search with a Holistic Combination Objective Score (HCOS) to search 2–5-drug regimens directly, implemented and tested against both synthetic and real patient data.
- **Nonsense-mediated-decay-aware TP53 resolution.** This pipeline distinguishes stable mutant protein from NMD-degraded truncating mutations using real TCGA pan-cancer evidence ^[24], validated on a real patient case, not only in the abstract.
- **A multi-agent adversarial check, not a single model's output.** Specialist reasoning agents debate candidate regimens, with disagreement surfaced explicitly rather than silently resolved.
- **Demonstrated self-correction.** This pipeline's own focal patient was found, via a real clinical-subtype cross-reference, to have been mis-classified as TNBC. The error, its detection method, and its correction are documented below as evidence of the verification discipline the approach is built on.

Preliminary Results

All results below come from an implemented, executed pipeline run against real Genomic Data Commons (GDC) TCGA-BRCA sequencing data — not simulated output.

A Corrected Focal Patient, and Why the Correction Matters

An initial focal patient (TCGA-A8-A08B) was selected via an automated 20-gene mutation scan and presented as an illustrative TNBC case. Cross-referencing against real PAM50 molecular subtype data later revealed this patient to be **BRCA_Her2** subtype, *not* TNBC. This was corrected by adding an explicit subtype-confirmation step (checking for **BRCA_Basal**, the standard PAM50 proxy for TNBC) before treating any patient as a TNBC case.

A Genuinely Confirmed TNBC Case, Found at Scale

171 confirmed-TNBC (BRCA_Basal) patients were cross-referenced against 992 locally available sequenced cases, yielding 168 patients with both confirmed subtype and real mutation data. Scanning all 168 through an expanded 90-gene RTK/NRTK panel yielded 93 patients with 2 or more altered genes.

Pipeline Output on the Selected Case

The focal case, **TCGA-AO-A128**, carries 11 altered genes, of which 6 (EGFR, PTEN, FLT1, TP53, ERBB2, TYK2) have a real drug mapping, yielding 15 candidate drugs. Running MDCOE's beam search, the top-ranked, toxicity-screened regimen was **afatinib (EGFR) + alpelisib (PI3K, via PTEN loss) + trastuzumab (HER2-directed)**, HCOS = 0.450. The ERBB2 finding is a missense *activating mutation without amplification*, supported by the Phase II SUMMIT trial's TNBC cohort of HER2-mutant, non-amplified patients treated with HER2-directed therapy ^[23].

A Second, Independently-Generated Hypothesis From the Same Patient

The same patient's TP53 nonsense mutation resolves, under the NMD-aware zygoticity logic developed for this proposal, to **adavosertib (WEE1i) + venetoclax (BCL-2i)** — a mechanistically distinct, p53-independent strategy, ranked lower under MDCOE's current scoring not for biological reasons but because the heuristic rewards shared-pathway overlap, and TP53's pathway does not overlap with the EGFR/PTEN/ERBB2 triplet's. This is a stated, testable limitation, not a hidden one.

Finding	What Was Done	Result
Subtype error caught	Cross-referenced focal patient against real PAM50 clinical data	TCGA-A8-A08B reclassified from TNBC to BRCA_Her2; pipeline corrected
Confirmed cohort built	Intersected 171 confirmed-TNBC patients against 992 sequenced cases	168 real, usable confirmed-TNBC patients with mutation data
Expanded gene panel applied	Scanned 168 patients through 90-gene RTK/NRTK panel	93 patients with 2+ altered genes; richest case had 11
Primary hypothesis generated	Ran MDCOE beam search + HCOS ranking on TCGA-AO-A128	afatinib+alpelisib+trastuzumab, HCOS=0.450, toxicity-clear
Secondary hypothesis generated	Applied NMD-aware TP53 zygosity resolution to same patient	adavosertib+venetoclax, mechanistically distinct, lower-ranked but not lower-confidence
Toxicity data gap found and closed	Audited all drugs in knowledge base against toxicity table	36 of ~50 drugs had no toxicity entry; all filled with real, label-based data before trusting results

Research Design and Methods

Aim 1 Methods: Computational Pipeline

Layer	Component	Function
L1	Multi-Omic Ingestion	Parses real GDC MAF files plus optional CNV data for amplification/deep-deletion detection.
L2	Knowledge Base	Currently a curated gene→pathway→drug→toxicity dictionary (103 genes, ~50 drugs); target architecture (KG+ +) extends this to a live graph.
L3	Synergy Scoring	A documented heuristic, explicitly not yet a trained model; Aim 3 proposes training this on DrugCombDB/AZ-DREAM.
L3.5	MDCOE Regimen Search	Beam search ranking 2–5-drug regimens by HCOS; implemented and tested on synthetic and real patient data.
L4	Multi-Agent Reasoning	Specialist agents debate top-ranked regimens; safety flags

Layer	Component	Function
		are surfaced, not silently discarded.

Aim 2 Methods: Wet-Lab Validation (Two-Arm Pilot)

Arm A: EGFR/PI3K/HER2-Directed Triplet

- Model systems: a PTEN-null, ERBB2-mutant (non-amplified) TNBC cell line or patient-derived organoid; MCF10A as non-malignant control.
- Design: fixed-ratio ray design (afatinib × alpelisib × trastuzumab) rather than a full 3D checkerboard.
- Readouts: viability, apoptosis, phospho-EGFR/AKT/HER2; cardiotoxicity monitored via primary cardiomyocytes.
- Comparator: afatinib+alpelisib alone (no trastuzumab), isolating the HER2-directed component's specific contribution.

Arm B: TP53-Directed Pairing

- Model systems: identical cell stock/organoid as Arm A — no new model system required.
- Design: standard 8×8 checkerboard (adavosertib × venetoclax).
- Readouts: viability, apoptosis, plus γH2AX foci and cleaved PARP.

Synergy quantification (both arms): Bliss independence, Loewe additivity, and HSA computed in parallel; a combination is only advanced as “confirmed synergistic” if ≥ 2 of 3 metrics agree.

Anticipated Results and Interpretation

The two-arm design creates four distinct outcome scenarios, and three of the four remain scientifically useful — not just the one where the model's top-ranked pick (Arm A) wins:

	Arm A synergistic	Arm A additive/antagonistic
Arm B synergistic	Both hypotheses validated — the strongest possible result, though least diagnostic for HCOS's ranking specifically, since both worked regardless of score.	Arm B wins despite its lower HCOS score — triggers the pathway-overlap-bias correction (Aim 3).
Arm B additive/antagonistic	Arm A validated, Arm B refuted — confirms HCOS's ranking was directionally correct; supports scaling MDCOE as-is.	Neither arm confirmed — an informative null result motivating Aim 1's GNN-replacement work more urgently.

Two further outcomes matter independent of the synergy call: **target engagement** (phospho-HER2 drop in Arm A; γH2AX rise in Arm B) can succeed or fail separately from viability — correct engagement with no net synergy would point to a downstream resistance mechanism rather than a wrong hypothesis — and **unpredicted**

toxicity, since the curated toxicity table is a label-based approximation, not measured in this specific cellular context.

A mild prior, stated explicitly so it can be checked against the data: Arm A's individual mechanisms each have independent trial or preclinical support, even though the triplet itself is untested; Arm B's WEE1i+BCL-2i logic is mechanistically defensible but has less precedent as a specific pairing. This is a weak expectation, not a prediction — the pilot is specifically designed to catch the case where this prior is wrong.

Aim 3 Methods: Closing the Loop

Wet-lab results from Aim 2 feed back into the pipeline via a seven-stage loop: (1) hypothesis extraction, (2) experimental model selection, (3) dose-response/synergy assays, (4) toxicity profiling, (5) mapping results back into the knowledge base and scoring model, (6) structured clinician/tumour-board feedback as an RL reward signal, (7) iteration across additional confirmed-TNBC patients. A specific, pre-registered test is built into this proposal: if Arm B (lower HCOS) matches or outperforms Arm A (higher HCOS) in wet-lab synergy, that is direct, actionable evidence that the pathway-overlap term in the current scoring heuristic is miscalibrated and should be down-weighted relative to evidence-strength and mechanistic-complementarity terms.

Timeline

Period	Activities
Months 1–3	Expand confirmed-TNBC cohort scanning to additional patients; begin GNN synergy model training on DrugCombDB/AZ-DREAM to replace the current heuristic (Aim 1).
Months 3–6	Execute the two-arm wet-lab pilot (Aim 2); complete toxicity and synergy data collection for both arms.
Months 6–8	Analyze wet-lab results against model predictions; apply the pre-registered pathway-overlap-bias test; update knowledge base and scoring weights accordingly (Aim 3).
Months 8–12	Extend validated pipeline to additional confirmed-TNBC patients from the existing 93-patient cohort; prioritise cases with no existing trial precedent.

Expected Outcomes and Impact

- A validated (or explicitly refuted) computational hypothesis for a real, confirmed TNBC patient, with either outcome being scientifically useful given the pre-registered comparison design above.
- Empirical calibration data on how well a heuristic, pathway-overlap-based synergy score agrees with real wet-lab synergy — directly informing whether and how to correct the scoring model before further scaling.
- A reusable pipeline and confirmed-TNBC patient cohort (93 patients, expandable) for generating and testing further patient-specific hypotheses, particularly for patients whose genomic profiles have no existing trial precedent.

- A concrete, demonstrated model for closed-loop refinement between computational hypothesis generation and wet-lab validation in precision oncology.

Future Architecture Extensions: A Phase 2 Roadmap

Scope note: everything below is proposed follow-on work, *not* part of Aims 1–3 and not yet implemented or tested. It is included to show a credible path beyond this proposal's funded scope, and to be explicit about which claims are validated-prototype (above) versus aspirational-roadmap (here) — a distinction this proposal treats as important throughout.

1. A Dual Synergy Engine: GNN + Causal Generative Model

The current heuristic rewards shared pathway overlap — a bias this proposal's own preliminary data exposed directly (TP53's pathway does not overlap with the EGFR/PTEN/ERBB2 triplet's, causing MDCOE to under-rank a mechanistically sound alternative). A trained GNN synergy predictor (SynerGNet/DeepDDS-style^[1,2]) addresses the *prediction* side of this; a causal component — e.g. a structural causal model combined with a GFlowNet-style generative policy over regimens — would additionally let the system learn counterfactual mechanistic complementarity (“if EGFR is inhibited and PTEN is lost, what pathway rewiring follows?”) rather than only pattern-matching pathway membership. This is a substantial research undertaking in its own right, requiring dedicated training data and compute beyond this proposal's Aims.

2. MDCOE++: From Beam Search to MCTS with a Learned Policy

Section 4.2.5 describes beam search or Monte Carlo Tree Search (MCTS) as conceptual alternatives; only beam search has been implemented and tested to date. Upgrading to MCTS with a learned policy network (guided by the GNN synergy engine) and value network (guided by the causal model above) would let regimen search improve directly from Aim 2's wet-lab outcomes rather than relying on a fixed heuristic — in effect, turning MDCOE from a static search into a reinforcement-learning-guided one. This directly targets the pathway-overlap bias this proposal identified, since a learned value function is not constrained to reward pathway overlap unless the data actually supports it.

3. KG++ With Multi-Layer Embeddings and TP53 NMD-State Edges

The current knowledge base is a static, hand-curated dictionary (Aim 1 Methods). A DREAMwalk-style multi-layer graph-embedding approach^[10] would let drug-gene-pathway-toxicity relationships, clinical-trial evidence, resistance mechanisms, and — as a genuine extension beyond DREAMwalk's published scope — this proposal's own NMD-aware TP53 zygosity distinction all be represented as learnable graph edges rather than rule-based lookups. This would make TP53-state-specific drug prediction a property the model learns and can refine from data, rather than a hand-written function.

4. A Counterfactual Mechanism Auditor Agent

A new addition to the multi-agent debate layer (Section 4.3): an agent that uses the causal model above to simulate alternative biological states (“what if this patient's TP53 were wild-type instead of nonsense-mutant?”) and flags regimens whose predicted benefit depends fragily on a specific, uncertain genomic call. This would directly strengthen Aim 3's feedback loop by testing not just whether a regimen scored well, but how robust that score is to plausible alternative readings of the same sequencing data.

5. A Toxicity-Prediction Transformer

Aim 1's preliminary data found 36 of ~50 drugs had no toxicity entry at all before this was manually corrected with label-based data. A transformer trained on FAERS (FDA Adverse Event Reporting System) and CTCAE (Common Terminology Criteria for Adverse Events) data, plus trial-report text, could predict *combination-specific*, not just single-agent, toxicity — directly relevant to Arm A's cardiotoxicity risk (trastuzumab), which the current label-based table can only approximate from single-agent data.

Revised Architecture (Text Diagram)

L1	Multi-Omic Ingestion	MAF + CNV + expression + TP53 NMD-state resolution
L2	Knowledge Graph (KG++)	DREAMwalk-style embeddings + toxicity + resistance edges
L3	Dual Synergy Engine	GNN synergy predictor + causal generative model
L3.5	MDCOE++	MCTS with learned policy (GNN) / value (causal) networks
L4	Multi-Agent Debate	+ new: Counterfactual Mechanism Auditor
L5	Closed Loop	wet-lab -> KG++ -> synergy engine -> MDCOE++ -> agents -> wet-lab

How This Would Change Aim 3

Under this roadmap, closing the loop (Aim 3) would extend from updating a scoring heuristic to: (a) updating causal-model weights directly from wet-lab outcomes, (b) retraining MDCOE++'s policy/value networks rather than re-tuning fixed formula weights, and (c) adding wet-lab-discovered resistance mechanisms as new KG++ edges rather than manual dictionary entries. Each of these is a meaningfully larger undertaking than the corresponding Aim 3 activity funded in this proposal, and would be the natural subject of a follow-on proposal once Aims 1–3 produce the wet-lab-validated calibration data this roadmap would need to train against.

Limitations

- The synergy-scoring heuristic is not yet a trained model; Aim 1's proposed GNN training is a prerequisite for scaling beyond hand-curated literature boosts.
- The knowledge base is a static, hand-curated dictionary, not the live-updating knowledge graph (KG++) the target architecture describes.
- Wet-lab validation in Aim 2 covers one patient's two hypotheses; broader validation across the 93-patient cohort is future work contingent on this pilot's results.
- Clinical covariates beyond PAM50 subtype (stage, prior treatment, comorbidities) are not yet incorporated into patient selection or scoring.

References

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Full technical detail — complete architecture diagrams, all 24 references, and the underlying tested codebase (extract_maf_patient.py, mdcoe.py, tnbc_combo_pipeline.ipynb, analyze_wetlab_results.py) — is available in the companion technical report and code repository.